

FULL GIT + GITHUB WORKFLOW GUIDE

This complete guide will walk you through the entire workflow used by developers when working with Git and GitHub.

1. Install Git

- Download from <https://git-scm.com/>
- Install with default settings

2. Configure Git (Important)

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@email.com"
```

3. Create Your Project Folder

```
mkdir myproject
```

```
cd myproject
```

4. Initialize Local Git Repository

```
git init
```

5. Check Status Anytime

```
git status
```

6. Add Files to Git

```
git add .
```

or add a single file:

```
git add app.py
```

7. Commit Changes

```
git commit -m "Initial commit"
```

8. Create a GitHub Repository

- Go to GitHub → New Repository → Create
- Do NOT initialize with README (recommended)

9. Connect Local Repository to GitHub

```
git remote add origin https://github.com/username/repo.git
```

10. Push Code to GitHub

```
git branch -M main
```

```
git push -u origin main
```

11. Daily Development Workflow

1. Edit code
2. git status
3. git add .
4. git commit -m "Explain changes"
5. git push

12. Pull Latest Changes (Very Important)

Always pull before new work:

```
git pull
```

13. Branch Workflow (Professional Way)

Create a branch:

```
git checkout -b feature-login
```

Work and commit:

```
git add .
```

```
git commit -m "Added login system"
```

Push branch:

```
git push -u origin feature-login
```

Open Pull Request on GitHub → Review → Merge

14. Switching Between Branches

```
git checkout main
```

```
git checkout feature-login
```

15. Merging Branches Locally

```
git checkout main
```

```
git merge feature-login
```

16. Cloning an Existing GitHub Project

```
git clone https://github.com/username/repo.git
```

17. .gitignore Rules

Essential Python ignores:

```
env/
```

```
__pycache__/
```

```
*.pyc
```

```
*.log
```

18. Undoing Mistakes

Undo staged files:

```
git reset
```

Undo last commit (keep changes):

```
git reset HEAD~1
```

19. View Commit History

`git log`

`git log --oneline`

20. Delete a Branch

Local:

`git branch -d branchname`

Remote:

`git push origin --delete branchname`

END OF FULL WORKFLOW GUIDE